

توثيق الاستثمار

Financial & Duration Estimation Report

Client: Confidential Client

Prepared By: Senior Software Engineering Management & Financial Estimation Team

Project Type: Enterprise Management & Data Processing Platform

Date: May 24, 2026

Angular FrontendJava Spring Boot BackendPostgreSQL DatabaseREST APISDockerAWS Cloud Infrastructure

1. Executive Summary

This report provides a professional enterprise-level financial and duration estimation for the development of a custom enterprise management platform based on the provided screen designs, screenshots, and functional descriptions.

The proposed solution includes: Complex enterprise data-entry screens, Dynamic CRUD operations, Video management functionality, Advanced filtering and search capabilities, Import/Export operations, Role-based access control, Reporting and audit logging, Enterprise deployment architecture.

Medium-High Enterprise Complexity Level

- Large number of data-entry forms
- Heavy business logic
- Advanced validation rules
- Multiple integrations
- Rich frontend interactions
- Complex backend processing

2. Project Overview

The project aims to modernize and centralize operational workflows through a scalable enterprise web platform that enables users to: Manage structured records, Process and retrieve complex data, Handle video-linked information, Perform advanced search operations, Export/import business data, Maintain centralized operational management.

3. Business Objectives

Objective	Expected Business Impact
Digitize manual operations	Reduce operational overhead
Centralize data management	Improve reporting accuracy
Improve user productivity	Faster workflows
Enhance scalability	Future extensibility
Improve auditability	Enterprise compliance

4. Scope Definition

✓ Included Scope

Enterprise Angular frontend, Spring Boot backend APIs, PostgreSQL database architecture, JWT Authentication, CRUD management modules, Video management module, Export/Import capabilities, Role-based authorization, CI/CD deployment pipelines, QA & testing, AWS deployment.

✗ Out of Scope (if any)

Mobile apps, Third-party payment gateways, Legacy data migration (to be discussed separately), Custom hardware integration.

5. Frontend Scope Analysis

Estimated Frontend Complexity: High

- A. Side Menu Module:** Multi-level navigation, 6 modules, Dynamic routing, Role-based visibility
- B. CRUD Screens (6 Screens):** Dropdowns, Dynamic forms, Data tables, Search & filtering, CRUD operations
- C. Major Retrieval Screens (3 Screens):** Advanced search logic, Dynamic filtering, Large data retrieval operations
- D. Medium Complexity Screens (2 Screens):** Simplified retrieval operations, Standard validations
- E. Highly Complex Enterprise Screens (2 Screens):** 30–40 fields each, Dynamic validation, Multiple sections, Advanced business logic, Complex UI state handling, Data grids
- F. Video Management Screen:** Video playback integration, Metadata storage, Dynamic table management, Record linking functionality
- G. Import/Export Features:** Excel, CSV, PDF formats

6. Backend Scope Analysis

Backend Modules	Complexity
Authentication & Authorization	Medium
CRUD APIs	Medium
Search & Filtering APIs	High
Video Metadata Processing	High
File Upload Management	High
Export/Import Processing	Medium-High
Audit Logging	Medium
Reporting Services	Medium
Database Optimization	High

7. Technical Complexity Assessment

Area	Complexity
Frontend Architecture	High
Backend APIs	High
Database Design	Medium-High
Security	Medium
Video Handling	High
Dynamic Forms	High
Deployment & DevOps	Medium
QA Coverage	High

Overall Project Complexity: **Medium-High Enterprise Complexity**

8. Duration Estimation

Phase	Duration
Discovery & Analysis	3 Weeks
UI/UX Design	4 Weeks
Frontend Development	12 Weeks
Backend Development	12 Weeks
QA & Testing	3 Weeks
Deployment & Stabilization	1 Week

Total Estimated Timeline: 35 Weeks (8 – 9 Months)

9. Financial Estimation

Human Resource Cost Breakdown

Category	Estimated Hours	Hourly Rate	Estimated Cost
UI/UX Design	160	\$10	\$1,600
Frontend Development	900	\$10	\$9,000
Backend Development	850	\$10	\$8,500
QA & Testing	300	\$10	\$3,000
DevOps & Deployment	120	\$10	\$1,200
Project Management	180	\$10	\$1,800

Infrastructure & Operational Costs

AWS Hosting (Basic Stable Setup)	\$500
PostgreSQL Hosting	\$300
CI/CD Pipelines	\$250
Monitoring & Logging	\$150
Backup & Security	\$200

Total Estimated Budget

Human Resources	\$25,100
Infrastructure	\$1,400
Contingency Reserve (10%)	\$2,650
Grand Total Estimated Budget	\$29,150

10. Timeline & Milestones

Milestone	Duration	Cumulative Week	Deliverable
Phase 1	Week 1–3	Week 3	Discovery & Analysis
Phase 2	Week 4–7	Week 7	UI/UX Design
Phase 3	Week 8–19	Week 19	Core Backend APIs
Phase 4	Week 8–31	Week 31	Frontend Development (parallel with backend)
Phase 5	Week 32–34	Week 34	QA & Integration Testing
Phase 6	Week 35	Week 35	Deployment & Production Release

11. Infrastructure & Deployment

Proposed Infrastructure

AWS EC2 Basic Instance, Docker Containers, CI/CD Pipelines, PostgreSQL Database, Secure REST APIs, Monitoring & Logging

Deployment Environments

Development, Staging, Production

12. Risks & Assumptions

Assumptions

- ✓ Requirements remain relatively stable
- ✓ Client provides timely feedback
- ✓ No major third-party integration changes
- ✓ Existing workflows are clearly defined

Risks

- ⚠ Scope expansion
- ⚠ Additional reporting requirements
- ⚠ Complex business-rule modifications
- ⚠ Infrastructure scaling changes
- ⚠ Additional security requirements

13. Payment Milestones

Milestone	Payment Percentage
Project Kickoff	20%
UI/UX Approval	15%
Backend Completion	25%
Frontend Completion	25%
QA & UAT Approval	10%
Production Deployment	5%

14. Final Recommendation

Based on the analyzed screenshots, system complexity, and enterprise architecture requirements, the project requires a structured development lifecycle while maintaining cost optimization and stable infrastructure services.

The proposed budget and timeline are optimized for:

- Stable enterprise-grade implementation
- Maintainable architecture
- Cost-efficient deployment
- Production-ready quality
- Scalable future enhancements

The project is technically feasible within the proposed optimized budget while preserving software quality and operational stability.